Living Data Management Plans with Data Stewardship Wizard

DSW

ds-wizard.org

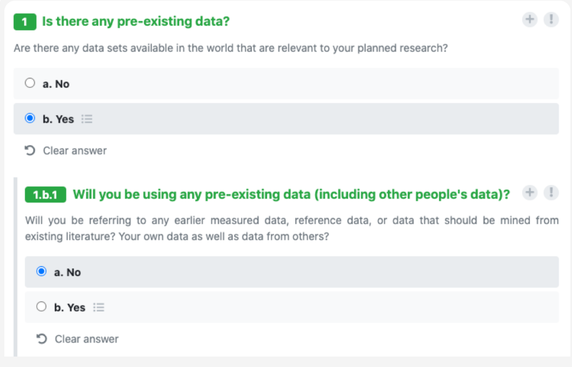

Fig.1 *Smart Questionnaire*

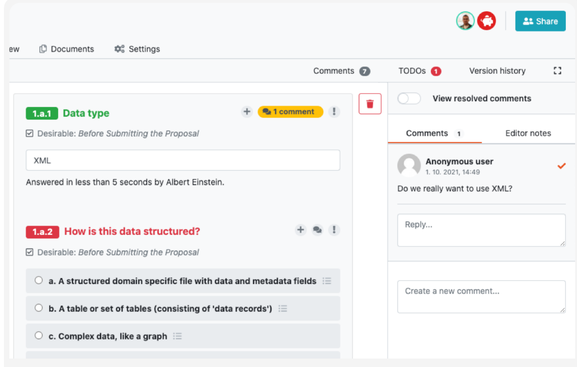

Fig.2 *Online Collaboration*

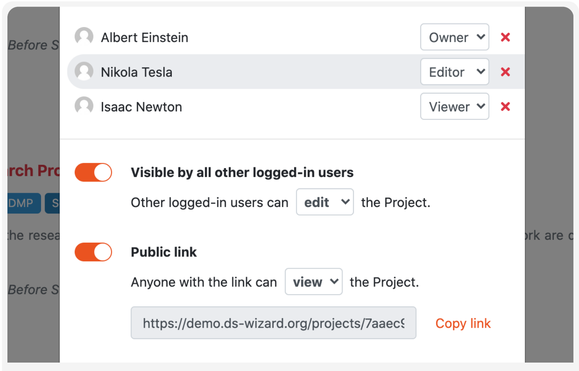

Fig.3 *Project Sharing Options*

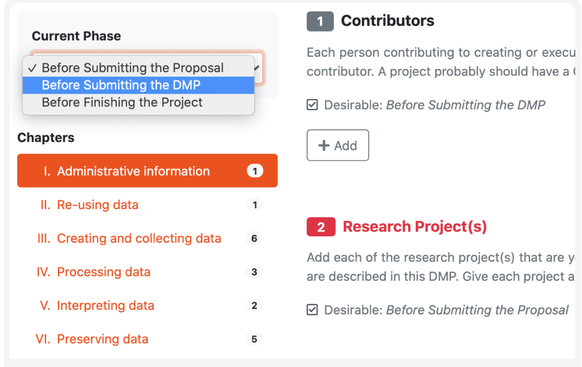

Fig.4 *Project Phases*

Researcher

Pick a Knowledge Model

A knowledge model is a template determining the structure of questionnaires. When creating a new project, you always choose the knowledge model that is the most suitable for you and your research. It is not bound to a specific DMP template but to a research field or institution.

Plan

DSW provides a simple way to create the DMP by filling in a smart questionnaire (Fig.1). It will guide you through the vast knowledge of data stewardship by asking questions and relevant follow-up questions, offering hints, external resources, FAIR metrics, and more.

Collaborate

Research is teamwork more often than not. In DSW, it is easy to share the project with your colleagues and collaborate on the DMP together online in real-time (Fig.2). Furthermore, you can set their access rights as needed and let them view, comment, edit or even own the project (Fig.3).

Get your data management plan

The knowledge models are not bound to a specific DMP template. So once you have your answers, you can choose what DMP template to use. Whether a Science Europe template or a machine-actionable DMP, you pick the DMP template and get your answers transformed into the desired format.

Maintain, Evolve

Things change, and it's essential to keep your project and DMPs updated. A project can have several phases that require different answers (Fig.4). You can improve and change answers and generate updated DMPs. Every change is saved into the version history so you can come back to any past version.

Knowledge Model

A knowledge model is a tree-like structure divided into chapters composed of questions. The questions include possible answers, references, and FAIR metric labels where relevant. There are also follow-up questions that depend on the previous answer. This way, you do not have to answer questions irrelevant to your research.

FAIR Metrics

The FAIR data recommendations are one of the steps in making data machine-actionable. According to these recommendations, data should be Findable, Accessible, Interoperable, and Reusable. It may not be easy to grasp the concept of FAIR fully, but DSW will make it easier for you to be comfortably up to date with the latest standards and best practices.

Integrations

DSW can be integrated with external services to get the answers while filling in the questionnaire. Then, once the DMP is generated, it can be submitted to external services using a submission service. Besides that, you can also use Single Sign-On to integrate DSW with your institution's system to share user accounts.

Document Templates

A document template describes how the replies from a questionnaire are composed into a document. This allows producing various documents from a single questionnaire using different templates. Each template can specify a number of formats such as human-readable PDF, Word, HTML, or machine-actionable RDF/XML, Turtle, or JSON-LD.

Incorporate FAIR Metrics

Some answers are better than others. In DSW, you can indicate how each answer affects the FAIR metrics to give hints for researchers on what to improve.

Integrate With External Services

It is possible to integrate questions with external resources or databases. Then, when researchers answer these questions, they can pick from an already existing list of answers. These answers also contain the link to the original resource.

Publish Knowledge Models for Use

When the knowledge model is ready to be used by researchers, you need to publish it first. This step involves assigning additional metadata such as a license or version number. Once the version is published, it is persistent and cannot be changed. You can always create a new version whenever you need.

Build Knowledge Models

The knowledge model editor (Fig.5) is a tool for building new knowledge models or extending the existing ones. Then they can cover new areas of expertise or better fit your organization's needs.

Create Document Templates

Anyone can develop a custom template that transforms answers from questionnaires in any way. There is no limitation; any textual representation can be created using the Jinja2 templating markup language. Template Development Kit (TDK) makes the development of custom templates easier.

Data Steward

DSW

DATA STEWARDSHIP WIZARD

Create, plan, collaborate, and bring your data management plans to life with a tool trusted by thousands of people worldwide — from data management pioneers, to international research institutes.

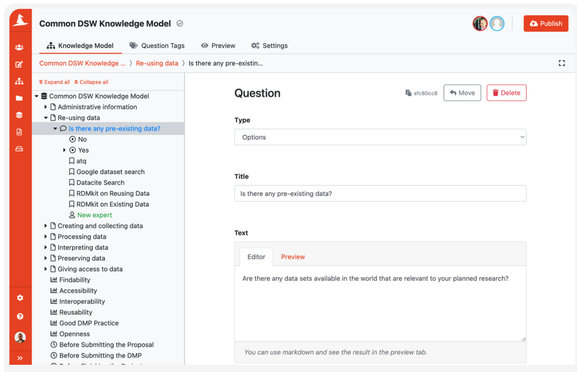

Fig.5 *Knowledge Model Editor*

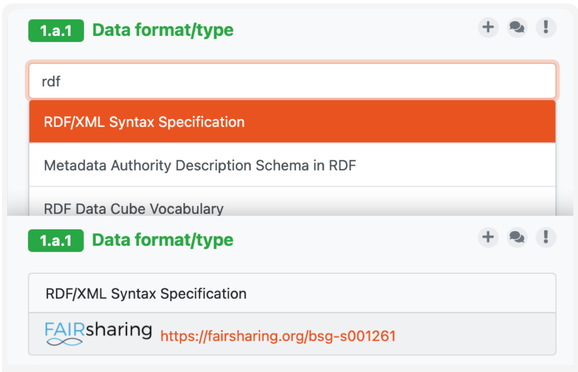

Fig.6 *Integration Question*

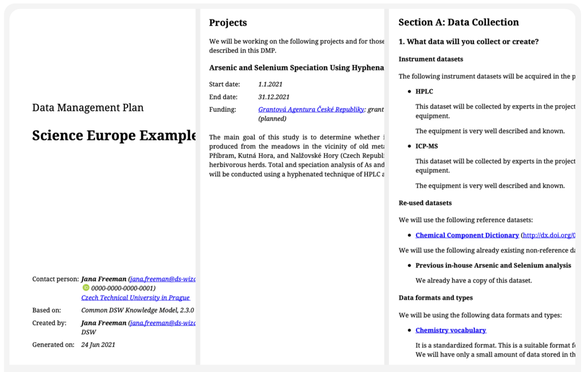

Fig.7 *Science Europe Template*

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